

Argument Principle.

Let f be a non-constant function, analytic at z_0 with $f(z_0) = 0$.

Then, for some $r > 0$, f have only z_0 as z_0 in $B_r(z_0)$. Thus,

$$f(z) = (z - z_0)^k \cdot F(z) \text{ for some } k \in \mathbb{N},$$

and analytic function F at z_0 with $F(z_0) \neq 0$. Now,

$$f'(z) = k(z - z_0)^{k-1} F(z) + (z - z_0)^k \cdot F'(z),$$

this implies
$$\frac{f'(z)}{f(z)} = \frac{k}{z - z_0} + \frac{F'(z)}{F(z)}.$$

Since $F(z_0) \neq 0$, f'/f has simple pole at z_0 , with residue k . So,

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i \cdot k \text{ where } C \text{ is simple closed curve around } z_0, \\ \text{is contained in } B_r(z_0).$$

Now, let f has a pole of order n at z_0 , then

$$f(z) = \frac{F(z)}{(z - z_0)^n}$$

where F is analytic at z_0 and $F(z_0) \neq 0$. Then,

$$f'(z) = -n \cdot \frac{F(z)}{(z - z_0)^{n+1}} + \frac{F'(z)}{(z - z_0)^n},$$

so therefore
$$\frac{f'(z)}{f(z)} = \frac{(z - z_0)^n}{F(z)} \cdot \left(-n \cdot \frac{F(z)}{(z - z_0)^{n+1}} + \frac{F'(z)}{(z - z_0)^n} \right) = \frac{-n}{z - z_0} + \frac{F'(z)}{F(z)}$$

so,
$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i \cdot (-n).$$

Obtain the two results, we obtain the following:

Thm. Let f be analytic inside and on a simple closed contour C except for finite poles in C , and $f(z) \neq 0$ on C . Then, counted with multiplicity, numbers of zeros N_f and poles P_f ,

$$\frac{1}{2\pi i} \cdot \int_C \frac{f'(z)}{f(z)} dz = N_f - P_f.$$

[More observation.

Suppose that $f(z)$ is analytic and non-zero for all z on a simple closed curve C .

Set $\log f(z) = \ln|f(z)| + i \cdot \arg f(z)$

where a fixed branch for the logarithm is chosen. Then,

$$\int_C \frac{f'(z)}{f(z)} dz = \int_C d(\log f(z)) = \ln|f(z)| \Big|_C + i \cdot \arg f(z) \Big|_C$$

By the fundamental theorem of line integrals, $\ln|f(z)| \Big|_C = 0$, so

$$\int_C \frac{f'(z)}{f(z)} dz = i \cdot \arg f(z) \Big|_C$$

Let C' be the image of C under f . There are two cases;

Thm. Rouché Theorem.

Let f and g be analytic inside and on a simple closed contour C , with
 $|g(z)| < |f(z)|$ for all $z \in C$.

Then, $f+g$ and f have the same number of zeros inside C .

First proof. By the hypothesis, $|g(z)| < |f(z)|$ on C . The triangle inequality gives

$$|f(z)| > 0 \quad \text{and} \quad |f(z) + g(z)| \geq |f(z)| - |g(z)| > 0 \quad (z \in C).$$

Therefore, f and $f+g$ are analytic inside and on C , and have no zeros on C .
By the analyticity, the number of poles, $P_f = P_{f+g} = 0$.

Since f and $f+g$ cannot be zero on the contour C , write

$$f(z) + g(z) = f(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right) = f(z) \cdot \phi(z) \quad \text{on } z \in C.$$

$$\begin{aligned} \text{Then, } (f+g)'(z) &= f'(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right) + f(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right)' \\ &= \frac{f'(z)}{f(z)} \cdot (f(z) + g(z)) + \left(1 + \frac{g(z)}{f(z)}\right)' \cdot (f(z) + g(z)) / \left(1 + \frac{g(z)}{f(z)}\right) \end{aligned}$$

$$\text{So that } \frac{(f+g)'(z)}{(f+g)(z)} = \frac{f'(z)}{f(z)} + \frac{\left(1 + \frac{g(z)}{f(z)}\right)'}{\left(1 + \frac{g(z)}{f(z)}\right)} = \frac{f'(z)}{f(z)} + \frac{\phi'(z)}{\phi(z)}.$$

Let N_{f+g} and N_f denote the number of zeros of $f+g$ and f , respectively, on the domain, bounded by C . By the argument principle,

$$N_{f+g} - N_f = \frac{1}{2\pi i} \cdot \int_C \left[\frac{(f+g)'(z)}{(f+g)(z)} - \frac{f'(z)}{f(z)} \right] dz = \frac{1}{2\pi i} \cdot \int_C \frac{\phi'(z)}{\phi(z)} dz. \quad (*)$$

Since $|g(z)| < |f(z)|$, $|g/f| < 1$ on C , so

$$|\phi(z) - 1| = |g(z)/f(z)| \leq 1.$$

Taking a branch of the logarithm containing $D = \{w \in \mathbb{C} \mid |w-1| < 1\}$, then

$$\frac{\phi'(z)}{\phi(z)} = \frac{d}{dz} \log \phi(z)$$

Therefore, the 'integral' in (*) is zero, by the fundamental theorem of line integral, consequently $N_{f+g} = N_f$. $\square$

Example.

Let $P(z) = z^{11} + 3z^{10} + 6z^3 + 1$. Determine the number of zeros of f in $\{z \in \mathbb{C} \mid 1 < |z| < 2\}$.

Proof. Let $f(z) = 3z^{10}$ and $g(z) = z^{11} + 6z^3 + 1$. By the triangle inequality, on $|z|=2$,

$$|g(z)| = |z^{11} + 6z^3 + 1| \leq |z|^{11} + 6 \cdot |z|^3 + 1 \leq 2^{11} + 6 \cdot 2^3 + 1 \leq 3 \cdot 2^{10} = 2^{11} + 2^{10} = |f(z)|$$

Therefore, by the Rouché's Theorem, $f+g$ has same number of zeros in $|z| < 2$ as f , 10 zeros. And now, let $f(z) = 6z^3$ and $g(z) = z^{11} + 3z^{10} + 1$. Observe that

$$|f(z)| = 6 \cdot |z|^3 \quad \text{and} \quad |g(z)| = |z^{11} + 3z^{10} + 1| \leq |z|^{11} + 3|z|^{10} + 1$$

We can choose $\delta > 0$ such that $0 < \epsilon < \delta$ implies

$$|1 + \epsilon|^{11} + 3 \cdot |1 + \epsilon|^{10} + 1 < 6.$$

Therefore, for $|z| = 1 + \epsilon$ where $0 < \epsilon < \delta$,

$$|f(z)| = 6 \cdot |1 + \epsilon|^3 \geq |g(z)|,$$

So $P(z)$ has three zeros inside $|z| = 1 + \epsilon$, for any $0 < \epsilon < \delta$.

Therefore, letting $\epsilon \rightarrow 0$, P has three zeros on $\{z \in \mathbb{C} \mid |z| \leq 1\}$, consequently it has 7 zeros in the annulus $1 < |z| < 2$. $\square$

✓ Evaluate

$$I = \int_{|z|=2} \frac{z+2}{z(z+1)} dz = \int_{|z|=2} \frac{\frac{1}{z^2} + \frac{2}{z^3}}{\frac{1}{z} + \frac{1}{z^2}} dz = - \int_{|z|=2} \frac{f(z)}{f'(z)} dz$$

where $f(z) = \frac{1}{z} + \frac{1}{z^2} = \frac{z+1}{z^2}$. It has order of 2 pole at $z=0$,
and has order of 1 zero at $z=-1$.

$$\text{Therefore, } I = -2\pi i \cdot (1-2) = 2\pi i.$$